

Component Attribution in Agentic Knowledge Graph Question Answering: A Controlled Ablation, Budget Analysis, and Failure Census

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Abstract

Large language models answer factual questions fluently, but they sound just as fluent when they are wrong. To fix this, researchers introduced retrieval-augmented generation, which grounds answers in retrieved text. But it retrieves only once, so a question spanning several facts is answered from that one query. Systems that navigate a knowledge graph do better. They interleave retrieval with reasoning, one hop at a time. Such a system has several parts. A planner breaks the question into steps. A scorer picks the next edge. A backtracker recovers from dead ends. A checker reads the draft answer. Papers report the system as a whole. So when it beats a baseline, nobody knows which part earned the gain. We built one, Agentic Graph Reasoning (AGR), with four removable parts, and took it apart. AGR and four baselines share one frozen backbone and one call budget on 400 questions each from WebQSP and ComplexWebQuestions. AGR leads the agentic baseline by 9.5 and 8.7 Hits@1 points. But the whole margin comes from questions where the shared budget cuts the baseline off. Where it finishes, the baseline is ahead. Removing the planner improves WebQSP F1 by 0.083 and cuts token use by 31%. The other three parts, claim verification among them, show no effect our tests could detect. A hand-read census of the failures finds label defects in 5.5% and 4.8% of the samples and names the *echo attractor*. It is a true entity one hop from the answer, so every grounding check passes it.

Keywords: knowledge graph question answering, large language models, agentic retrieval, ablation study, component attribution, benchmark quality

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1. Introduction

Large language models answer factual questions fluently, and often enough they answer them wrongly. A fabricated answer reads in the same register as a correct one, so it is hard to catch from the output alone [1], and the problem compounds once an answer requires several facts put together: if the first hop lands on the wrong entity, nothing later in the chain will know. Retrieval-augmented generation [2] grounds the model in retrieved text, but it retrieves once, before any reasoning starts, and the passage that holds an intermediate entity need not resemble the question at all. A knowledge graph [3] turns composition into traversal, since each hop is an edge, and static graph retrieval [4] still fetches a neighbourhood in one shot. Agentic systems close that gap by interleaving retrieval with reasoning: the model takes one step, looks at what it found, and decides where to look next [5, 6, 7, 8].

These systems are assemblies. A typical one decomposes the question into sub-goals, scores candidate edges with the model, backtracks from dead ends, and checks its own output, and the literature reports the assembly as a whole. When it outperforms a static baseline, three things could have produced the gain: a component, the budget the comparison ran under, or the labels of the benchmark itself. The report rarely separates them, so later work inherits the whole assembly on faith, along with its cost. The exception shows why taking a system apart matters: KBQA-o1’s own ablation removes its tree search, and GrailQA F1 falls by thirty points [8]. This paper does the same for an agentic knowledge-graph question-answering system built so that each of its components can be removed cleanly, and it asks which of the three sources produced the gain.

The system is **Agentic Graph Reasoning** (AGR), an explicit state machine over a constrained graph tool API in which a program runs the loop and the language model is consulted only at bounded decision points. Four components can be removed: a *planner*, an *edge scorer*, a *backtracker*, and a *structural verification layer* that splits each draft answer into atomic claims and checks them against the traversed triples before the answer is emitted. The last is the component the system was designed around, and its two graph routes ask whether an edge joins a claim’s endpoints, not whether it is the edge the claim asserts, so a claim that X is Y’s mother is certified by an edge recording that X is Y’s child. Sections 3.2 and 8.1 bound it further.

On 400 questions from each of WebQSP [9] and ComplexWebQuestions [10], over a Freebase-derived environment of 2.59 million entities and 8.31 million triples, AGR reaches Hits@1 of 0.755

and 0.522 and leads four baselines that share its backbone and its call budget. That is where the analysis starts, not where it ends. Against a reimplementations of Think-on-Graph [5] on the same tools, the entire margin comes from the 29% and 44% of questions on which the shared 25-call cap truncates the baseline; where the cap lets it finish, it is ahead (Section 5.3). The ablation finds one component with a detectable effect, and the effect runs the wrong way: removing the planner improves WebQSP F1 by 0.083 while cutting tokens by 31%, because decomposing a question that has no compositional structure throws away the context that would have answered it. Backtracking, model scoring, and claim verification show no effect at the power available, and Section 6.3 states the smallest effect each test could have caught. The failure census finds label defects on 5.5% and 4.8% of the sampled questions, and names a failure that passes every grounding check and that consensus-based label repair cannot see.

2. Related Work

Three lines of work bear on question answering over knowledge graphs: what retrieval fixed before reasoning can reach, what interleaving retrieval with reasoning buys, and what checking an answer against evidence adds. None establishes which component of an agentic system is responsible for its behaviour, and that gap is the one this paper addresses.

Retrieval-augmented generation [2] embeds the question, retrieves similar passages, and answers from them. Where the answer must be composed across passages the ranking signal measures the wrong thing, because the passage carrying the intermediate entity resembles neither the question nor anything the system yet knows to look for; surveys of the graph–language-model intersection identify this as the central limitation of flat retrieval [11]. Knowledge graphs make the missing structure explicit, so composition becomes a traversal: GraphRAG [4] retrieves structured context in place of passages, and a second family keeps retrieval static but learns it, scoring candidate subgraphs with a graph neural network [12] or retrieving a bounded subgraph with a light-weight scorer [13]. All of them retrieve once, before reasoning, so a fixed-radius neighbourhood is either too small to contain the answer or so large that the relevant edges are lost among thousands of irrelevant ones.

Interleaving retrieval with reasoning removes that constraint. Think-on-Graph [5] established the pattern with model-guided beam search over relations and entities, and its successor adds

document retrieval at each step [14]. Plan-on-Graph [6] added decomposition into sub-objectives and reflection-driven backtracking; AGR’s planner and backtracker are those two mechanisms re-hosted in a program-orchestrated loop, its verification layer is the addition, and Section 8.2 records that Plan-on-Graph itself is not among the systems run here. Reasoning-on-Graphs (RoG) [7] instead generates grounded relation paths as explicit plans with a fine-tuned model and traverses them, so answers are faithful by construction, as does graph-constrained reasoning [15]; RoG is also the system whose published subgraphs this paper’s environment is built from (Section 4.1), and the supplementary material compares the two directly. KBQA-o1 [8] is closest to AGR’s search strategy, performing agentic knowledge base question answering with full Monte Carlo Tree Search over a ReAct-style [16] agent. AGR does none of this: its search is guided best-first with a bounded beam and backtracking, carrying neither rollouts nor value backpropagation.

A parallel line checks generated content after the fact. Chain-of-Verification [17] has a model draft a response, plan verification questions, answer those independently and regenerate, checking against the model’s own knowledge rather than an external symbolic source, whose limits are documented [18]. A second group critiques the *retrieval*, emitting reflection tokens that judge whether a passage supports the output [19] or grading retrieved documents and triggering a fallback search [20], while FActScore [21] decomposes generations into atomic facts as a metric rather than a control step. The layer of Section 3.2 sits between these lines: it decomposes a draft into claims and checks them against an external source, but the source is the set of triples the agent itself traversed and the check runs before the answer is emitted. The path-faithful systems [7, 15] make the question moot by never generating a claim the graph did not produce. To our knowledge no graph-navigating agent that generates its answer freely checks that answer’s claims against its own traversal before emission, and whether that check earns its place is what Section 6 measures.

The gap that motivates this paper is therefore methodological. Agentic systems are assemblies, and the literature reports them as wholes, so a published gain over a static baseline rarely establishes which mechanism produced it, whether some contribute nothing, or how much is the budget regime the comparison ran under. The exception is instructive: KBQA-o1’s own ablation removes MCTS while leaving every other component intact, and GrailQA F1 drops from 78.5 to 48.5 [8] — a thirty-point attribution to a single mechanism, obtained precisely because the authors ablated. Section 6 applies the same instrument to AGR, and Section 5.3 adds the second source of a reported

gain, the budget, which an ablation cannot see.

3. The Instrument: AGR

AGR inverts the usual agent pattern. It does not hand the model a set of tools through native function-calling and let the model drive the loop. A program orchestrates every step, calls the graph tools deterministically, and consults the model only at bounded decision points, through prompts constrained to return JSON. Every graph operation is therefore logged by construction rather than reconstructed from the model’s account of what it did, and budgets are enforced by counters a router checks rather than requested in a prompt the model may ignore. The consequence this paper uses is that each of the loop’s four components can be removed by a configuration flag without touching the others, which is what Section 6 does; the supplementary material states exactly what each flag removes.

3.1. State Machine

A state-machine diagram in the supplementary material gives the topology: six agent nodes over a shared typed state, with three cycles. The **Planner** decomposes the question into an ordered chain of at most four sub-objectives in one model call, from a prompt whose first exemplar is a single-hop question that is *not* decomposed. **Explorer** and **Evaluator** form a search loop: the explorer expands the frontier by scoring every candidate relation of at most five anchors, blending embedding similarity against the active sub-objective with a batched model judgement at a frozen $\alpha = 0.7$, and the evaluator judges whether that objective is satisfied from the thirty most similar traversed triples. It may accept only entities the graph returned. The **Backtracker** restores an earlier, higher-scoring frontier and bans the edges that led away from it; that ban list is what turns backtracking into a search operation, since without it a deterministic scorer facing the same restored frontier would select the same edges again. The supplementary material gives the four graph operations, the scoring rule, and the router and the backtracker’s snapshot discipline. Every path from Evaluator to Answerer passes through the **Verifier**, and the budgets recorded in the supplementary material bound all three cycles, so no sequence of model outputs can produce a non-terminating run.

3.2. Structural Verification Layer

Nothing in the navigation loop constrains what the final answer *asserts*. The verifier is the check that does, and it is the component the system was designed around. One model call drafts prose from the traversed triples, extracts the entities the draft asserts as its answer, and decomposes the draft into atomic claims $\langle h, r, t \rangle$ over entity *names*, since the model has never seen a graph identifier. Each claim is then routed through up to three tests and leaves at the first that settles it: **traversed adjacency**, which asks whether the claim’s endpoints were walked together and is the only route of the three that attaches citable evidence; **verify_connection**, which asks the full graph the same question for endpoints that were reached but never walked together; and, for whatever clears neither, an **entailment check** against the fact window the drafter saw. Symbols decide, and the model is consulted only where the graph is silent. If no claim is unsupported the draft is emitted verbatim; otherwise the verifier either sends the agent back to explore for the missing support or, when the budget is spent, rewrites the draft around a deterministically filtered entity list. The supplementary material gives the routing and the three outcomes in full.

What the graph routes actually check bounds every claim made for this layer. Both of them ignore the relation and the direction, so what they establish is that the endpoints of a claim are connected in the environment, not that the claimed relation is the one the edge records: a claim that X is Y’s mother is certified by an edge recording that X is Y’s child. Only one of the three routes attaches evidence at all. Section 8.1 states the exposure that follows and how much of it the committed record can bound, and Section 6.2 measures what the layer changes.

The layer is also nearly free, which matters because a cost argument is sometimes offered in place of an accuracy one. The grounded path adds nothing beyond the single drafting call already required, and at test scale removing the layer altogether saves 2% and 6% of tokens on the two ablation half-splits (Table 3). Whatever the case for or against it, it is not a cost argument.

4. Experimental Setup

Everything in this section exists to make one inference safe: that a difference between two systems is a difference between architectures. Where something is *not* held constant, we name it here.

4.1. Knowledge Environment

Every dataset this paper consumes is public, standard, and third-party, and all of it arrives through one distribution: the per-question subgraphs published with Reasoning-on-Graphs [7] as rmanluo/RoG-webqsp and rmanluo/RoG-cwq, built over a Freebase [3] snapshot and used unmodified. What is built here is the environment derived from them: a property graph of 2.59 million entities and 8.31 million triples over 7,058 distinct relation types, the union of every per-question subgraph across both datasets and all three splits. All five systems navigate that one graph, just under two-thirds of whose nodes (63.7%) are unnamed mediators encoding n -ary facts.

An environment assembled this way is friendlier than full Freebase, and we state the consequence rather than leave it to be inferred. Before any system ran, a validation gate measured what fraction of questions have at least one gold answer reachable from a topic entity within the four-hop cap: 97.0% on the WebQSP sample and 99.2% on the ComplexWebQuestions sample. A sample that passed that gate is what the paper calls *certified*. That ceiling is why a failure in Section 7 can be attributed to the system rather than to a missing edge, and equally why **absolute accuracies here do not transfer to a full-Freebase setting**: every comparison in Section 5 is between systems traversing the identical environment, which is what the comparison is for. A second limit concerns expressiveness: the environment carries no typed literal and the tool API no ordering operator, so a question answered by a date, a quantity, or a superlative is unanswerable here by construction.

4.2. Benchmarks, Baselines, and Backbone

We evaluate on WebQSP [9] and ComplexWebQuestions [10], the two standard multi-hop KGQA benchmarks over Freebase. **We sample 400 questions from each** rather than running the full 1,628 and 3,531 test splits, a pre-specified fallback adopted when the full-split cost was projected at roughly three times the budget available. The sample is stratified by hop count at a fixed seed, so its strata are proportional to the split’s: 256, 128, and 4 questions at one, two, and three or more hops on WebQSP, with 12 whose gold answer the gate found unreachable, and 137, 211, and 49 on ComplexWebQuestions, with 3 unreachable. The resulting 95% bootstrap intervals are roughly ± 4 to ± 5 points on Hits@1. The benchmarks differ less in the typical question than in the tail. Exactly half of the WebQSP sample carries a single gold answer, but its mean is 7.15 gold entities per question and its largest question lists 237, while ComplexWebQuestions averages 1.75 with a median of 1; that tail is one reason we report F1 beside Hits@1.

Four baselines span the design space, three of them controls that bound what a paradigm can do here rather than tuned competitors. A **no-retrieval** control answers from parametric memory alone, and **Vector-RAG** retrieves verbalised triples by embedding similarity over a sentence encoder [22]. **Static GraphRAG** [4] retrieves a fixed-radius neighbourhood in one shot; its radius is one logical hop and its expansion is capped and unordered (detailed in the supplementary material), so it licenses one narrow claim and no more: *retrieval fixed before reasoning must commit to a radius in advance, and any fixed radius is wrong for some question in the set*. **Think-on-Graph** [5] is the agentic comparator and the one comparison here between two systems of the same paradigm: model-guided beam search over relations and entities, reimplemented on this environment’s tools at the original paper’s beam width and depth of 3 and 3 (detailed in the supplementary material). Reimplementing holds graph, model, budget, and scoring rule constant so that only the algorithm varies, and it also means its numbers are not comparable to the published ones.

Every system shares one frozen backbone, gpt-5.4-mini-2026-03-17, a dated snapshot rather than an alias, at temperature 0.0 with reasoning effort disabled, together with the entity resolver, the question files, the output schema, and *one budget meter with the same 25-call cap*; both graph-navigating systems seed from the datasets’ annotated topic entities, which removes entity linking as a confound. Holding all of this constant is what licenses the paper’s central inference: a difference between two systems is attributable to architecture, not to model capacity, retrieval corpus, or spend. Three things are deliberately not held constant. The baselines receive a plain answering prompt, while AGR’s drafting prompt carries the grounding provisions that are part of the system under test. The two navigators consider candidate sets of different widths, 40 and 20 against 300 and 200. And the shared call cap binds on them at very different rates. The second is where a reader should look first for a confound, and Section 5.3 reports the measurement that bounds it.

4.3. Metrics and Protocol

Accuracy is reported as **Hits@1** and per-question **F1**. All five systems return an unranked set of answer entities, so Hits@1 follows the benchmarks’ convention for that case: a question scores a hit if any returned entity is in the gold set. F1 sits beside it because a system can score a hit on that rule while asserting several wrong entities alongside the right one. A **hedge** is an empty predicted entity set; hedges score zero on both metrics and the hedge rate is its own column, so

a system buying precision through abstention is visible rather than flattered. Hits, hedges, and wrong answers partition the whole set, and nothing is dropped from the denominator. Matching is exact after Unicode normalisation, case folding, and whitespace stripping; of AGR’s wrong answers, 1 of 65 on WebQSP and 6 of 99 on ComplexWebQuestions are surface-form near-misses that aggressive normalisation would have scored as hits, so the census of Section 7 is reading real errors. **Groundedness** is measured at two levels, one deterministic over every asserted entity and one a sampled judgement of whether an entity is supported *as the answer* (both defined in the supplementary material). That judge was validated against a human annotator on 100 items, reaching 85% agreement and Cohen’s $\kappa = 0.6995$. That *misses* the 0.70 we pre-specified, by 0.0005, and Section 8 treats the miss as a limitation rather than rounding it away. Efficiency is tokens and model calls, both exact from the shared meter.

The protocol was **pre-specified**: research questions, metrics, baseline certification thresholds, and ablation conditions were fixed and documented before the test sets were constructed, and no hyperparameter was tuned on test. Four decisions were written down before the measurements they govern — the frozen configuration $\alpha = 0.7$ and $\tau = 0.20$ of Section 3, selected on a disjoint 80-question development set; the sample size of 400 questions per dataset; the baseline certification threshold; and the judge threshold of $\kappa \geq 0.7$. The last was missed, and it is reported as missed, because a threshold that counts only when it is met is not a threshold. Nothing was filed with a public registry, so the claim is pre-specification and not pre-registration. Confidence intervals are 95% percentile bootstrap intervals over questions from 10,000 resamples, and paired comparisons use the exact form of McNemar’s test on per-question correctness. The thirteen prompt templates the five systems send, AGR’s seven and the baselines’ six, are reproduced verbatim in the supplementary material, generated from the code by a script so that they cannot drift.

5. Aggregate Results and Where the Margin Lives

The same runs are read several times over: what the five systems answer correctly and what they spend, then where the margin over the agentic baseline comes from, and whether the entities they assert are supported by the graph they navigated. Where AGR is not the best of the five, it says that too.

Table 1: Main results, 400 questions per dataset, one frozen backbone and one 25-call budget for every system. Hedge is the proportion of questions answered with an empty entity set. The three single-call systems are controls that bound a paradigm; the comparison the paper reads is the one between the two navigators.

System	WebQSP				ComplexWebQuestions			
	Hits@1	F1	Hedge	Calls	Hits@1	F1	Hedge	Calls
no-retrieval	0.453	0.305	12.2%	1.0	0.307	0.279	38.0%	1.0
Vector-RAG	0.568	0.432	27.0%	1.0	0.203	0.168	48.2%	1.0
GraphRAG	0.340	0.262	55.8%	1.0	0.205	0.183	60.8%	1.0
Think-on-Graph	0.660	0.477	18.8%	12.8	0.435	0.378	22.5%	18.2
AGR	0.755	0.642	8.2%	6.2	0.522	0.469	23.0%	8.9

5.1. Accuracy

Table 1 gives the main matrix. AGR has the highest Hits@1 and F1 in every column, reaching Hits@1 of 0.755 on WebQSP against the agentic baseline’s 0.660 and 0.522 on ComplexWebQuestions against 0.435, so the margin over Think-on-Graph is 9.5 and 8.7 points. The per-question F1 gap is wider still, 0.642 against 0.477 and 0.469 against 0.378, because F1 penalises the extra wrong entities that a set-intersection hit forgives. All eight paired comparisons against the four baselines reject at the conventional level. The two against Think-on-Graph, the only comparisons in which both systems navigate the same graph with the same model under the same call budget, rest on 76 questions AGR alone answers against 38 the other way on WebQSP ($p = 4.8 \times 10^{-4}$), and 86 against 51 on ComplexWebQuestions ($p = 3.5 \times 10^{-3}$). Against RoG’s published figures, measured over the full splits with a backbone fine-tuned on both benchmarks, AGR trails on all four; the supplementary material gives that comparison and what it does and does not establish.

Two features cut against a simple reading. **Raw hits mislead on the static graph baseline.** GraphRAG scores 0.340 on WebQSP, below the no-retrieval control’s 0.453, which invites the conclusion that it is broken; the wrong-answer counts invert the picture. The control asserts on 351 of 400 questions and is wrong on 170 of them, an *assertion precision* of 51.6%, where assertion precision is the share of questions a system commits on that it answers correctly. GraphRAG asserts on 177 and is wrong on 41, or 76.8%, because the retrieval baselines answer only from the facts retrieved and abstain when retrieval misses while the control guesses from memory. A grounded system can score below an unconstrained one on raw hits while committing four times

fewer falsehoods.

And AGR does not lead on every column. On ComplexWebQuestions its hedge rate of 23.0% is marginally *above* Think-on-Graph’s 22.5%, so the accuracy lead is not bought by answering more freely. The two systems commit at effectively the same rate, on 308 and 310 of 400 questions, and from those near-identical commitments AGR is right 209 times and wrong 99, against 174 and 136 for Think-on-Graph: 67.9% assertion precision against 56.1%. It finds 35 more answers and asserts 37 fewer falsehoods than beam search does, from the same number of commitments.

5.2. Cost

AGR reaches those numbers at roughly half the model calls of the agentic baseline, 6.2 against 12.8 on WebQSP and 8.9 against 18.2 on ComplexWebQuestions, ratios of 0.48 and 0.49. The call saving is genuine and the token saving is not: AGR spends 4,511 mean tokens per question on WebQSP against Think-on-Graph’s 3,615, and 6,818 against 5,572 on ComplexWebQuestions. It is the *number of round trips* that halves, not the volume of text, so on tokens Think-on-Graph is a frontier point rather than a dominated one, while on calls it is dominated on both datasets — and calls are what the shared budget meters. Wall-clock time tells the same story in seconds (reported in the supplementary material). AGR never reaches the shared call cap: across all 800 test questions it hit the 25-call limit zero times, though its depth cap refuses a deeper expansion on 25.6% of test questions. The baseline reaches the cap far more often.

5.3. Where the Margin Comes From

Beam search costs width times depth, so the shared 25-call cap binds on Think-on-Graph far more often than on AGR. At width 3 a full depth costs the baseline up to thirteen calls (Section 4.2), and the reimplementation reached its third expansion on 6 of 400 WebQSP questions and 11 of 400 on ComplexWebQuestions, so its published depth of three is nominal under this cap. Splitting the questions by whether the cap cut it off separates two things the aggregate confounds: how good its search is, and whether it can afford to finish. Table 2 gives the split, and it reverses the comparison on one side. Where the budget lets Think-on-Graph run to completion it is *ahead* of AGR, 0.852 against 0.788 on WebQSP and 0.629 against 0.607 on ComplexWebQuestions. The entire aggregate margin comes from the 29% and 44% of questions on which the cap truncates it, where its accuracy collapses to 0.197 and 0.188 while AGR still reaches 0.675 and 0.415.

Table 2: AGR against Think-on-Graph, split on whether the shared call cap truncated the baseline. Read from the per-record budget_exhausted trace flag.

Subset	n	Think-on-Graph Hits@1	AGR Hits@1
<i>WebQSP</i>			
Think-on-Graph finished	283	0.852	0.788
Think-on-Graph clipped	117	0.197	0.675
<i>ComplexWebQuestions</i>			
Think-on-Graph finished	224	0.629	0.607
Think-on-Graph clipped	176	0.188	0.415

The split conditions on the baseline’s own outcome, so the reading would be unsafe if the clipped half were simply the hard half, and the supplementary material reads it within hop strata, and every figure the reading rests on is quoted here. On WebQSP the clipped half is not harder for AGR: 81 of the 117 clipped questions are single-hop, AGR scores 0.728 on them against 0.766 on the single-hop questions the baseline finished, and the baseline’s fall from 0.846 to 0.160 across the same two sets is the cap, not the questions. On ComplexWebQuestions the halves do differ — AGR scores 0.309 on the clipped single-hop questions against 0.609 on the finished ones — so there truncation and difficulty are confounded and the affordability reading is weaker. The finished halves are not uniform either: on ComplexWebQuestions the baseline leads AGR only on single-hop questions, 0.754 against 0.609, and trails on two-hop, 0.561 against 0.610, which is the planner asymmetry of Section 6.1 seen from the other side.

The correct reading is affordability, not resolution quality. Exhaustive beam search finds answers well when allowed to finish; guided search finds them almost as well and finishes within a budget the beam search cannot. On a fixed spend the second is the better architecture; the first would win if spend were unconstrained. The cap of 25 calls is itself a choice, and a sweep of it is the first experiment Section 8.4 names.

One asymmetry bounds how far that reading extends, and it is the candidate-width difference named in Section 4.2. Both navigators call the same tools but truncate what those tools return at different widths: Think-on-Graph keeps the first 40 relations per entity and the first 20 neighbours per relation, against AGR’s 300 and 200. The relation list is sorted by *descending* fanout, so a binding cut discards the low-fanout tail, which is where the relation answering a specific question

tends to sit. Over the committed tool logs the 40-relation cut binds on 31.6% of the 1,651 entities Think-on-Graph expanded and the 20-neighbour cut on 32.8% of its 7,992 neighbour calls, against 1 of AGR’s 3,097 expanded entities and 3.3% of its neighbour calls.

The affordability conclusion survives this, because a thinner candidate pool cannot explain why a system runs *out of calls*. What it bounds is the other half of the split: the 0.852 and 0.629 the baseline reaches where it finishes were measured over a narrower pool than AGR’s, so they are a **lower bound** on what an equal-width reimplementation would reach rather than an estimate of it, and since the baseline is already ahead there, equalising the widths would be expected to widen that lead. Equalising them is the first change we would make in a follow-up. That re-run would not vary one thing, though: wider candidate sets make each pruning call dearer, so equal-width pruning would also raise the baseline’s clip rate and move the boundary of the very split it would be read from. The two effects have to be read apart, which means reporting that split at each setting rather than an aggregate in which they cancel or compound invisibly. The supplementary material adds the check on whether AGR buys hits by naming more entities; it does not.

5.4. Groundedness

Across both datasets AGR asserted 1,709 entities, and **zero of them were ungrounded**: every entity it named exists in the environment and is reachable from a topic entity within the hop cap, while the parametric control asserted 1,001 entities of which 22.1% failed that test. Two qualifications come first. The property is close to guaranteed by construction for any system that copies its entities out of tool results. And **it belongs to graph navigation, not to the verification layer**: Think-on-Graph also reaches exactly zero ungrounded assertions, on 1,265 entities, without any claim-checking component whatsoever.

A second tier asks whether an asserted entity is supported *as the answer*, and it separates the systems far less; the supplementary material gives every cell, and no between-system gap there survives $n = 60$. The level is the useful finding rather than the ranking. **Zero structural hallucination coexists with between a third and nearly two-thirds of assertions failing a strict semantic-support test**, for every system, including the two that are structurally perfect. Semantic support is hard for the paradigm, not for one system within it.

Table 3: Component ablations. $\Delta F1$ is ablated minus reference: positive means the system got better without the component. The interval is a paired percentile bootstrap over questions (10,000 resamples); p from McNemar’s exact test on paired per-question correctness.

Removed	WebQSP ($n = 200$)				CWQ ($n = 198$)			
	$\Delta F1$	95% CI	p	Tokens	$\Delta F1$	95% CI	p	Tokens
Planner	+0.083	[+0.038, +0.130]	0.006	−31%	−0.047	[−0.108, +0.013]	0.088	−21%
Backtracking	−0.013	[−0.038, +0.010]	0.727	−7%	0.000	[−0.031, +0.030]	1.000	−18%
Verification	+0.004	[−0.006, +0.018]	1.000	−2%	−0.009	[−0.024, +0.000]	1.000	−6%
Model scoring	−0.016	[−0.060, +0.026]	0.664	−25%	−0.008	[−0.046, +0.030]	0.481	−31%

6. Component-Level Attribution

Four conditions each remove exactly one component from the unmodified system, as the supplementary material specifies, and re-run the identical questions: no planner, no backtracking, no claim verification, and embedding-only scoring ($\alpha = 1$). Each is compared against the reference on the same half-split, drawn as half of every hop stratum rounded down, which gives $n = 200$ on WebQSP and $n = 198$ on ComplexWebQuestions, with McNemar’s test on the paired per-question outcomes. The reference row is the unmodified system restricted to the same questions, read from the main-run records rather than re-run, so it is provably the system that produced Table 1. **Throughout this section a delta is ablated minus reference, so a positive number means removing the component *improved* the metric.** The p -values in Table 3 are reported as computed; a correction for multiple comparisons was not among the pre-specified decisions, and applying one after seeing the results would be a choice made on the results, so the four tests are read as descriptive.

6.1. When Planning Hurts

One component produced a detectable effect, and it produced it in the wrong direction. **Removing the planner improves WebQSP F1 by 0.083 ($p = 0.006$)** while cutting token use by 31%. Twenty-one questions were answered correctly only without the planner, against six correct only with it, so the cheaper system is the more accurate one.

The mechanism is over-decomposition. WebQSP is predominantly a single-hop benchmark: 256 of its 400 questions need one hop. Asked to decompose a question with no compositional structure, the planner manufactures sub-objectives that are not really separable, the explorer then

scores candidate edges against a fragment of the question rather than the question, and it discards exactly the context that would have identified the right relation. The planner prompt carries a single-hop exemplar to show that not decomposing is a valid output, and that is evidently not enough. The per-stratum breakdown locates the damage: on the WebQSP half-split, removing the planner raises one-hop Hits@1 from 0.75 to 0.85 and two-hop Hits@1 from 0.84 to 0.89.

On ComplexWebQuestions the direction reverses, as the mechanism predicts. That benchmark is genuinely compositional, only 137 of 400 questions are single-hop, and removing the planner costs 0.047 F1. **That arm does not reach significance** ($p = 0.088$), and we report it as a direction rather than an effect; we do not call it marginally significant, because at this sample size that phrase would assert more than the data carries. The strata show a coherent mechanism rather than noise: without the planner, one-hop Hits@1 *improves* from 0.49 to 0.54, two-hop collapses from 0.48 to 0.34, and three-hop-plus moves from 0.62 to 0.54. That is the pattern in which decomposition helps genuine composition and hurts questions that do not need it, and it is the pattern the hop-strata table in the supplementary material shows from the other side. Pooling the two datasets would average a significant improvement against a non-significant harm and report approximately nothing. The asymmetry *is* the result: decomposition is not a general-purpose benefit but a mechanism whose value depends on whether the question has compositional structure to find.

6.2. Three Components with No Detected Effect

The remaining three conditions moved nothing detectable, and we report them at the same length as the one that did. A field that reports only what moved cannot attribute anything.

Backtracking. Removing it changes WebQSP F1 by -0.013 ($p = 0.727$) and ComplexWebQuestions F1 by 0.000 ($p = 1.000$), while saving 7% and 18% of tokens. The mechanism fires rarely: the reference condition records 38 and 108 backtracks over the 398 paired questions, and on those same runs the meter refuses a further attempt on 28 of them, 7.0%. A component this seldom exercised cannot show a large effect. One implementation detail bounds the null further and is recorded rather than smoothed over: the backtracker bans the edges of the most recent pass but restores an often older snapshot, so the null describes backtracking as implemented rather than as specified (detailed in the supplementary material). The accurate summary is that the search rarely needs to undo a decision, not that undoing is worthless.

Model scoring. Setting $\alpha = 1$ removes the language model from edge selection entirely. F1

falls by 0.016 and 0.008, neither significant, while tokens drop 25% and 31%. The model term is the single most expensive part of the navigation loop, and this experiment cannot establish that it earns its cost. The development-set sweep that froze α points the same way at a smaller scale, and one confound goes with the condition; The supplementary material gives both.

Claim verification. Removing the checking routes of the layer AGR was designed around, while leaving the drafting call unchanged, changes WebQSP F1 by +0.004 and ComplexWebQuestions F1 by -0.009 , $p = 1.000$ on both. Across the half-splits — 200 questions on WebQSP and 198 on ComplexWebQuestions — the two conditions disagreed on exactly one question each. **Claim verification does not measurably change how often AGR is right.** The result is expected rather than disappointing, for the reason Section 3.2 gave before any of these numbers appeared: on most questions the layer is a pass-through certifying a draft that was already grounded. On test it asks for a repair on 52 of 800 questions, exploration actually resumes on 28, and 13 end grounded that did not start so. A mechanism that alters the output on a few per cent of questions cannot move an aggregate accuracy metric, and an ablation is the right instrument for establishing that rather than assuming it either way. What the layer delivers is therefore not an accuracy gain but the output contract: an answer paired with the traversed triples that support it, and an entity list that is a deterministic function of the verdicts rather than a second generation. That is an auditability claim, not an accuracy claim, and Section 8.1 states how far it reaches; Section 5.4 has already shown that the layer cannot claim credit for groundedness.

6.3. Statistical Power

These are nulls at low power, not demonstrations of absence. McNemar’s exact test sees only the *discordant* pairs, and the null conditions produced very few. Removing claim verification changed the outcome on exactly one question per dataset, and with a single discordant pair no split whatsoever can reach significance, so that row is not a test that failed to reject: it is the two conditions agreeing on 396 of the 398 paired questions, a statement about how rarely the layer alters an answer rather than about statistical power. Backtracking produced 8 and 11 discordant pairs, and model scoring 21 and 18. At those counts the smallest imbalance the exact test could have called significant is 8 and 9 questions for backtracking, meaning all eight would have had to fall the same way, and 11 and 10 for model scoring, which is between 4.0 and 5.5 points of accuracy. Differences below that were undetectable at any power, whatever their true size.

For calibration: detecting a true 2:1 discordance ratio at 80% power requires roughly 72 discordant pairs, and the largest of these conditions produced 21, at which count the smallest ratio detectable at 80% power is nearer 4:1, so a true 2:1 effect would have been caught about a quarter of the time. **Three of four components show no effect at this power**; none is shown to have no effect. One more count bounds what any single removal could have done: the union of the four discordant sets holds 42 of the 200 WebQSP questions and 55 of the 198 on ComplexWebQuestions. On the remaining 158 and 143, no condition changed the outcome. The intervals in Table 3 say the same thing in F1, since the planner’s WebQSP interval is the only one of the eight that does not contain zero. Each condition was also run once and the components were removed one at a time, so three nulls are consistent with the three being individually dispensable and jointly necessary, and equally with a system that has none of them matching this one at a fraction of the tokens; the condition that separates those readings was not run, and Section 8.4 names it.

7. Error Analysis

After the label-defect exclusions of Section 7.2, AGR’s residual non-hits on the two samples number 262: 86 on WebQSP and 176 on ComplexWebQuestions. Six of them, one and five, are the surface-form near-misses of Section 4.3, scoring artefacts rather than reasoning failures, and were set aside. Every one of the remaining 256, 85 and 171, was read and labelled by hand against its full execution trace, so the counts are over a population rather than estimates from a sample. The reading protocol was fixed before it began: for each trace, find where the trajectory *first* left the correct path, and label that point rather than the last visible symptom. Wrong answers and hedges are kept separate, because a wrong answer is a reasoning error while a hedge is usually a coverage gap that never produced a committal claim. A histogram in the supplementary material gives the census by category. Its largest categories are unsurprising — wrong relation choice is the base rate of graph navigation, and composite claims dominate ComplexWebQuestions, whose template-generated questions carry conjunctions that WebQSP’s search-log questions rarely do. Questions whose answer is a date, a number, or a superlative fall into the class Section 4.1 named as unanswerable here, and on ComplexWebQuestions the system hedges on them more than three times as often as it asserts something wrong, 24 against 7. Two smaller categories are reported in full instead, because they generalise past this architecture.

7.1. The Echo Attractor

The most instructive failure mode is one that every grounding check in this paper passes. The system resolves a real entity, reached through a real edge, and states a fact that is genuinely true. It is not what was asked. We name this the **echo attractor**: a true, well-grounded entity sitting one hop from the answer in the graph’s structure. It takes two shapes. *Topic confusion* picks a neighbour in a tight naming cluster, answering that John Glover plays Lionel Luthor when asked who plays Lex Luthor. *Granularity* picks the right entity family at the wrong level, answering “Kingdom of Denmark” where the gold answer is “Denmark”, a genuinely distinct node. A reader may prefer to call the second a label-granularity problem; we count it against the system because the two nodes are distinct in the graph and the question’s gold names one of them, and the label-defect pass of Section 7.2 is where the other reading was tested.

What makes this a mechanism rather than a residue category is that nothing went wrong in the parts of the system built to prevent things going wrong. The right edge exists, is retrieved, and is walked; the entity returned is present in the environment, and every structural check certifies it. The failure is entirely in *which* true fact gets surfaced, which a claim-level check against traversed adjacency cannot catch by construction, because the claim is true.

The consequence for evaluation methodology is larger than the consequence for this system. Systems that share a graph and a backbone converge on the *same* wrong entity for the *same* questions, and Section 7.2 measures how often, so any protocol treating systems as independent observers cannot see it. Consensus filtering, ensembling, and majority-vote gold repair all take agreement as evidence of correctness, and here agreement is evidence of a shared bias instead, so such a protocol will confidently ratify the wrong answer in exactly the cases it was introduced to catch. The echo attractor accounts for 13 of the 256 failures read. The rate is small and the sample does not support a precise one: the contribution is the named mechanism and its methodological consequence, not its frequency at this n .

7.2. Benchmark Label Defects

The census required a second pass, because a failure against a wrong gold label is not a failure of the system. A consensus pre-pass flagged every question where at least three of the five systems agreed on an answer the benchmark scored wrong, and each was then adjudicated individually against the graph and the question text. The two counts must be kept apart. The pre-pass **flagged** 58

WebQSP and 47 ComplexWebQuestions questions; adjudication **confirmed** a genuine label defect in 22 and 19 of them, which is 5.5% and 4.8% of each sample, splitting into gold answers that are simply wrong (12 and 17) and questions whose phrasing admits several defensible answers of which the benchmark lists one (10 and 2). All 41 were removed before the census began, as benchmark artifacts rather than system failures. The hand-read then found 16 more that the pre-pass had missed — 3 on WebQSP and 13 on ComplexWebQuestions — so 57 **distinct questions** across the two benchmarks turned out to need correcting. That the census recovered defects a dedicated consensus pass had missed is itself an argument for reading failures individually rather than screening them in aggregate.

The gap between flagged and confirmed is not adjudication noise. The pass flagged 105 questions and adjudication confirmed 41, so roughly three-fifths of what it flagged was several systems converging on the same wrong entity. The supplementary material reads those 64 cleared questions row by row; the triage filed 52 of the 85 cleared rows as the echo mechanism of Section 7.1 and the rest as relation confusion, an ignored constraint, or the environment gap. Those rows also show that the agreeing systems are not independent observers for a second reason besides the graph: they share one backbone and one resolver, and the no-retrieval control, which sees no graph at all, joins the majority of the cleared ComplexWebQuestions rows (shown in the supplementary material). Convergence there is at least as much a property of what the backbone believes as of the graph’s neighbourhood structure, and the methodological consequence does not depend on which cause dominates: had we accepted the pre-pass without per-item adjudication, we would have “corrected” the benchmark toward the systems’ shared bias and reported an inflated accuracy for every system, including our own.

The accuracy tables are not rescored. Section 5 reports every system exactly as measured over the full samples. Rescoring against labels we adjudicated ourselves would substitute one contestable ground truth for another, and the defects cut both ways, so removing the adjudicated questions moves every system by at most 0.020 Hits@1 and changes no ranking (recomputed in the supplementary material). The adjudication was made by one author with the consensus signal visible, and Section 8.3 carries that as the limitation it is.

8. Discussion and Limitations

The limits of this study bound its claims, and we state all of them rather than the flattering subset.

8.1. *What Bounds the Instrument*

Wrongful acceptance is unmeasured, and it is the most serious gap in this work. The verifier logs the claims it rejects and not the ones it accepts, so a claim certified on an edge unrelated to the relation it asserts leaves no trace. The exposure is real rather than hypothetical, because both structural routes ignore the relation and the direction: any edge between two entities certifies any claimed relation between them. Nothing in the committed record says how often that happened, and the supplementary material gives the interval, two orders of magnitude wide, that is all the record supports. The same gap means the layer has no measured precision or recall as a classifier of claims, which is the first number a verification component should carry.

The output contract cannot be audited from the run records. AGR emits an answer paired with the traversed triples supporting it, and the pairing holds inside the run, but the logger writes the *count* of supporting triples and discards the list, so no committed artifact in this work contains a single supporting triple. Persisting each accepted claim with its route and its matching triples is one logging change, and it would repair this limitation and the previous one together.

Structural grounding is not answer adequacy. Both navigating systems assert zero ungrounded entities, which under the definition used is close to guaranteed for any system that copies entities out of tool results. It says their answers are made of the graph’s own material; it does not say they answer the question. The echo attractor of Section 7.1 is the counter-example, a wrong answer that passes every check this paper applies.

8.2. *What Bounds the Comparisons*

Sample size, and one budget. Every result rests on 400 questions per dataset rather than the full splits, giving 95% bootstrap intervals of roughly ± 4 to ± 5 points on Hits@1. The main accuracy margins clear that width, at 9.5 and 8.7 points, as does the planner effect, and the groundedness contrast at zero against 22.1% is not close; the semantic-tier gaps, the hedge-rate difference on ComplexWebQuestions and the last step of the rising hop-depth curve are within noise and reported as such. The 25-call cap was likewise pre-specified but is still a choice, and Section 5.3 shows the

entire margin over the agentic baseline living on the questions where it truncates. A cap sweep would move that boundary; it was not run.

An identical environment is not identical access to it. The agentic baseline truncates candidate sets at 40 relations and 20 neighbours against AGR’s 300 and 200, a cut binding on about a third of its expansions and effectively none of ours (Section 5.3), which is why its figures where it finishes are a lower bound rather than an estimate. It is also the only agentic comparator: Plan-on-Graph, whose two mechanisms AGR’s planner and backtracker re-host (Section 2), was not run, so the planner result speaks to decomposition as implemented here and to Plan-on-Graph’s by analogy only. **The static graph baseline is radius-bounded** in turn, so its decay across hop strata measures the radius as much as the paradigm, and the supplementary material declines to offer it as evidence for either.

One backbone, and nondeterminism. All five systems run on gpt-5.4-mini, frozen and closed, so this paper establishes how these architectures compare *on one backbone*, not how the comparison scales across model capacities. Temperature-zero decoding against a hosted API is greedy rather than deterministic, and measured trajectory stability is about 67%. Each system and each ablation condition was run once and no seeds were averaged, so the intervals and paired tests bound the sampling variance of the question draw and nothing else. Large defended margins and per-question pairing make a rank reversal from decoding noise unlikely, but unlikely is the strongest statement available: the repeated matrix that would bound the variance was affordable at the observed cost and was not run, which is a gap in the design rather than a budget constraint.

Ablation power. The four ablations run on half-splits of $n = 200$ and $n = 198$, and the paired test sees only the questions on which two conditions disagree, as few as one for claim verification. Three of four components show *no effect detected at this power*, which is not the same as no effect, and Section 6.3 gives the smallest difference each could have detected. The components were also removed one at a time, so the three nulls do not say whether a system with none of them would match this one.

8.3. What Bounds the Measurements

The environment is friendlier than the knowledge base it stands in for, and its node identity is coarser. Per-question subgraphs with a 97%-plus answer-reachability ceiling are a more forgiving substrate than full Freebase: between-system comparisons hold, because all five systems navigate

the identical graph, but absolute accuracies do not transfer, and the supplementary material shows the gap to a published system trained on the benchmarks. Topic entities are **given, not linked**, which isolates navigation from entity linking as intended but means every reported accuracy assumes a linking step a deployed system would have to perform. And entities are keyed by name, so **homonyms merge**: distinct real-world entities sharing a surface form collapse to one node, which can manufacture a path between things not actually connected and can equally destroy a distinction a question depends on. Neither direction is measured.

The measured accuracy has an unmeasured floor. An extraction defect converts correct reasoning into a wrong entity list on one question shape: on “list the professions” and “what did *X* do” questions the drafted text names every gold value and the scored entity list collapses to the sentence’s subject. Accuracy on that shape is a lower bound rather than an estimate.

The judge missed its own bar, and the adjudication was single-annotator. Agreement between the semantic-tier judge and a human annotator reached $\kappa = 0.6995$ against a pre-specified 0.70 — a miss, by 0.0005, which is why those numbers are reported as corroborating rather than decisive and why no conclusion rests on them alone. The judge also runs on the same backbone as the systems it judges. Separately, the benchmark-defect exclusions and the failure census rest on single-annotator judgements made by an author with the consensus signal visible (Section 7), and one development-set relabelling was made after its outcome was known. It is flagged in place, and no aggregate was rescored.

Scope. English factoid questions over a static graph, with no temporal facts, no graph construction, and an offline prototype. The environment carries no typed literals and the tool API no ordering operator, so questions answered by a date, a quantity, or a superlative are out of reach by construction (Section 4.1). Wall-clock figures are magnitudes with a cache caveat (reported in the supplementary material), not a production-readiness claim.

8.4. Outlook

The findings fix the order of the next experiments, and each is cheap against the cost of the study it would complete: the agentic baseline at AGR’s candidate widths under a sweep of the call cap, replication with the joint-removal condition Section 6.3 names, the full test splits, and one logging change that would convert the two limitations opening this section into measurable quantities. The supplementary material says what each would settle.

The most actionable finding points forward rather than back. If decomposition helps only when a question has compositional structure, improving accuracy by 0.083 F1 where that structure is absent and costing 0.047 where it is present, then the planner should not run unconditionally: **gate decomposition on question structure**. A hop-count predictor gating the planner would capture both arms of that asymmetry, and it can be evaluated with the instrument used here. More generally, agentic components should be treated as conditionally rather than universally beneficial, and the condition is worth measuring.

9. Conclusion

We built an agentic knowledge-graph question-answering system whose four components can be removed one at a time, ran it against four baselines on one frozen backbone under one call budget, and took it apart. The aggregate result is the one the field usually reports: the agentic system has the highest accuracy in every column, at roughly half the model calls of the agentic baseline. The rest of the paper is what that result turns out to be made of. The entire margin over the agentic baseline comes from the questions on which the shared budget truncates it, so it is a finding about affordability under a fixed spend rather than about reasoning power. Of four components, three could not be shown to earn their cost at the power available — including the claim-verification layer the system was designed around, which changes the answer on one question per dataset and delivers an output contract rather than an accuracy gain. The fourth was actively harmful on the shallower benchmark: removing the planner improved accuracy and cut token use, because decomposing a question that has no compositional structure discards the context that would have answered it.

The shape of the residue matters as much as the margin. A system that traverses a real graph and copies its answers out of real triples can be made to assert nothing ungrounded, and the navigation paradigm achieves that, not any checking layer built on top. What remains is an answer that is grounded, connected, and true, and is still not the answer to the question that was asked; systems sharing a graph and a backbone converge on that answer together, so a protocol that takes their agreement as evidence ratifies it. Fact verification against a knowledge graph turns out to be the easier half of the problem, and relevance verification is the half still open. The attribution result is finally a claim about how this field measures rather than about this architecture: a component adopted because the assembly containing it performed well may be contributing nothing, and a

margin reported as an architectural advantage may be a budget regime. The only way to know is to remove the component, vary the budget, and test.

CRedit authorship contribution statement

Md. Sakif Khan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Sadia Sharmin:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

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Data availability

Code, prompts, per-question run records, and the analysis scripts that produce every figure and table in this paper are available at <https://github.com/sakif-khan/agent-ic-graph-reasoning> under the MIT licence. The question sets are the public WebQSP [9] and ComplexWebQuestions [10] releases, and both they and the Freebase-derived [3] subgraphs arrive through the public Reasoning-on-Graphs distribution [7] (rmanluo/RoG-webqsp and rmanluo/RoG-cwq), used unmodified. The knowledge environment is rebuilt from that distribution by the scripts in the repository rather than redistributed.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to draft and revise the prose of this manuscript from the completed thesis on which it is based, and to write the scripts

that check the values quoted in the text against the study’s recorded results. The study design, the experiments, the analysis, and the findings reported here are the authors’ own and predate the preparation of this manuscript. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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